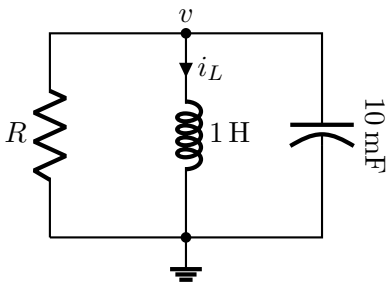
EE2015: Electric Circuits and Networks

Tutorial 7

(October 3, 2025)

1.

For the circuit shown on the right it is given that $v(0) = 5\,V$ and $i_L(0^-) = 0$. Find $v(t)$ for $t \geq 0$ when (i) $R = 1.923\,\Omega$, (ii) $R = 5\,\Omega$, and (iii) $R = 6.25\,\Omega$.



2. The responses of a series RLC circuit are, for $t \geq 0$, given by

$$v_C(t) = 30 - 10\,e^{-20t} + 30\,e^{-10t}\,\text{V}$$
$$i_L(t) = 40\,e^{-20t} - 60\,e^{-10t}\,\text{mA}$$

Determine the values of R, L, and C that are consistent with the given expressions.

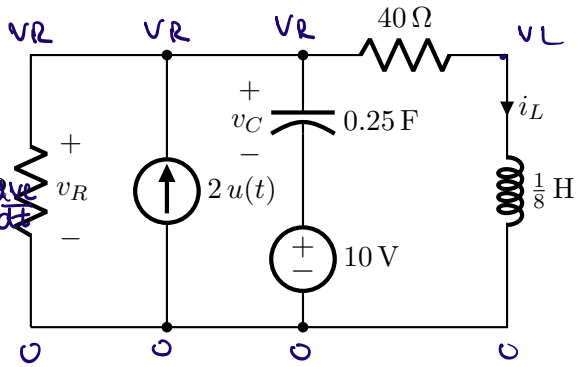
3.

For the circuit shown on the right find the following:

(a) $i_L(0^+)$, $v_C(0^+)$, $v_R(0^+)$

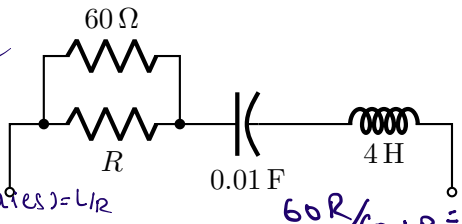
(b) $i_L'(0^+)$, $v_C'(0^+)$, $v_R'(0^+)$

(c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$



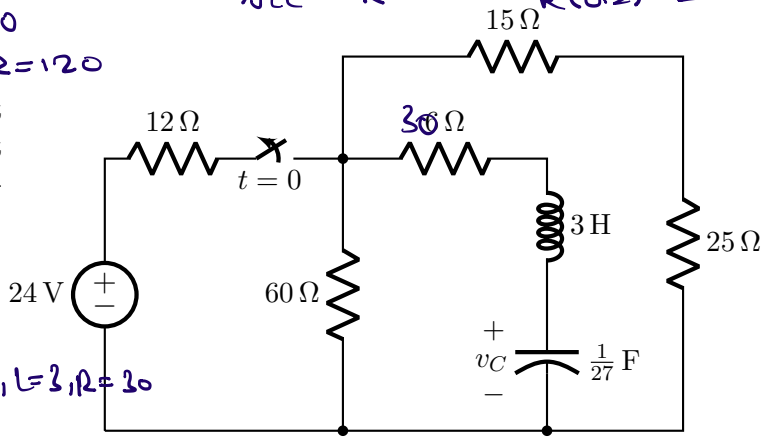
4.

For the circuit shown on the left, find the value of R such that will give rise to a crtically damped response.



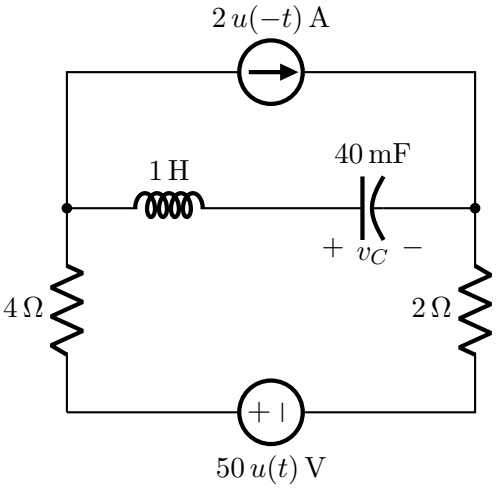
5.

For the circuit shown on the right find $v_C(t)$ for $t \geq 0$. Assume that the switch has been in the closed position for a long time prior to $t = 0$.



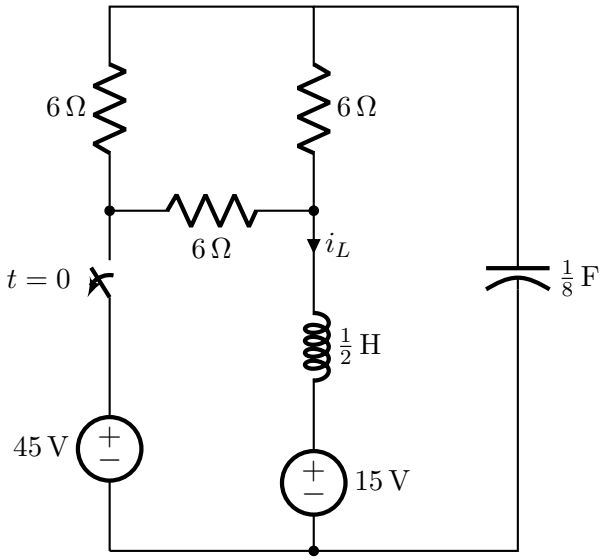
$v_C(0) = -12\,\text{V}$

6.



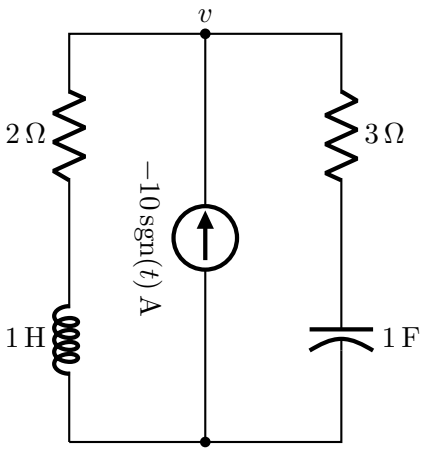
For the circuit shown on the right find $v_C(t)$ for $t \geq 0$.

7.



For the circuit shown on the left, find $i_L(t)$ for $t \geq 0$.

8.

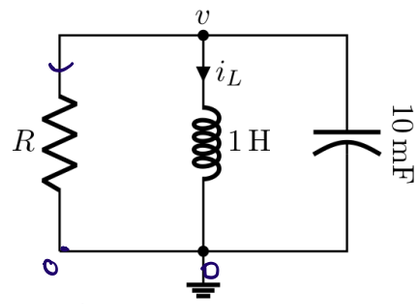


Find $v(t)$ for $t \geq 0$.

(October 5, 2025)

1.

For the circuit shown on the right it is given that $v(0) = 5\,V$ and $i_L(0^-) = 0$. Find $v(t)$ for $t \geq 0$ when (i) $R = 1.923\,\Omega$, (ii) $R = 5\,\Omega$, and (iii) $R = 6.25\,\Omega$.



$v/R + i_L + C\,dv/dt = 0$

$\Rightarrow C\frac{d^2v}{dt^2} + \frac{dv}{dt}(1/R) + v/L = 0$

$\Rightarrow \frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{1}{LC}(v) = 0$

$s^2 + 2\alpha s + \beta = 0$

$v/R + C\cdot v'(0) = 0$

$\Rightarrow v'(0) = -\frac{500}{R}$

$\omega_0 = 1/\sqrt{LC} = 10$

$2\alpha = 1/RC \Rightarrow \alpha = \frac{50}{R}$

$\alpha > \omega_0 \Rightarrow$ overdamped

$\alpha < \omega_0 \rightarrow$ underdamped

$\alpha = \omega_0 \Rightarrow$ critically damped

(i) $R = 1.923 \Rightarrow \alpha = 26 \Rightarrow$ roots are $-26 \pm \sqrt{26^2 - 10^2} = -26 \pm 24 = -2, -50$

$v(t) = A_1 e^{-2t} + A_2 e^{-50t}$

$A_1 + A_2 = 5$

$-2A_1 - 50A_2 = -500/1.923 = -260 \Rightarrow A_1, A_2$

(ii) $R = 5 \Rightarrow \alpha = 10 \Rightarrow$ roots are $s = -10$

$v(t) = (A_1 + A_2 t)\,e^{-10t}$

$\boxed{A_1 = 5}$

(iii) $R = 6.25 \Rightarrow \alpha = 8 \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2} = 6$

$v(t) = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$

$= e^{-8t}(A_1 \cos 6t + A_2 \sin 6t)$